

A Boundary-Trace Counterexample to the Power Law for Optimal Fractional Poincaré Constants

A full counterexample to Conjecture 5.8 of arXiv:2203.02425

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Abstract

Railo and Zimmermann conjecture that every optimal fractional Poincaré constant is a power of any one such constant. We disprove this already for $p = 2$ on the interval $(0, \pi)$. The optimal order-one constant is $C_{1,0} = 1$, attained by the zero extension of $\sin x$. At order two, the space of zero extensions has the additional clamped trace $u' = 0$ at both endpoints. Compactness and the equality case of the sharp Poincaré inequality then give $C_{2,0} < 1$. Thus $C_{2,0} \neq C_{1,0}^2$.

Status. full counterexample; likely valid. Human review is recommended.

1 The conjecture

For an open set $\Omega \subset \mathbb{R}^n$, the source uses

$$\tilde{H}^{r,p}(\Omega) = \overline{C_c^\infty(\Omega)}^{H^{r,p}(\mathbb{R}^n)}.$$

For $p = 2$, write $C_{r,z}(\Omega)$ for the optimal constant in

$$\|(-\Delta)^{z/2} u\|_{L^2(\mathbb{R}^n)} \leq C_{r,z}(\Omega) \|(-\Delta)^{r/2} u\|_{L^2(\mathbb{R}^n)}, \quad u \in \tilde{H}^r(\Omega). \quad (1)$$

Conjecture 5.8 asserts that, whenever $r > z \geq 0$ and $t > s \geq 0$,

$$C_{t,s}(\Omega) = C_{r,z}(\Omega)^{(t-s)/(r-z)}. \quad (2)$$

Theorem 2.3 makes us to suggest the following conjecture on the equivalence of the fractional Poincaré constants.

Conjecture 5.8 (Equivalence of the optimal fractional Poincaré constants). *Let $\Omega \subset \mathbb{R}^n$ be an open (bounded) domain and $1 < p < \infty$. If $C_{r,z}$ is the optimal fractional Poincaré constant for $r > z \geq 0$, then $C_{t,s} = C_{r,z}^{\frac{t-s}{r-z}}$ is the optimal Poincaré constant for $t > s \geq 0$.*

2 The interval and its traces

Take

$$n = 1, \quad p = 2, \quad \Omega = (0, \pi).$$

Functions are identified with their zero extensions to $\mathbb{R}$. Standard one-dimensional trace theory gives

$$\begin{aligned} \tilde{H}^1(0, \pi) &= H_0^1(0, \pi), \\ \tilde{H}^2(0, \pi) &= H_0^2(0, \pi) = \{u \in H^2(0, \pi) : u(0) = u(\pi) = u'(0) = u'(\pi) = 0\}, \end{aligned}$$

where the equalities mean zero extension into the corresponding space on $\mathbb{R}$. Plancherel gives

$$\|(-\Delta)^{1/2}u\|_2 = \|u'\|_2, \quad \|(-\Delta)u\|_2 = \|u''\|_2 \quad (3)$$

on these spaces.

The extra order-two trace is the decisive point: the zero extension of $\sin x$ belongs to $\tilde{H}^1(0, \pi)$ but not to $\tilde{H}^2(0, \pi)$.

3 The order-one optimum

The sharp Dirichlet Poincaré inequality on $(0, \pi)$ is

$$\|u\|_2 \leq \|u'\|_2 \quad (u \in H_0^1(0, \pi)),$$

with equality exactly for scalar multiples of $\sin x$. The zero extension of $\sin x$ is an admissible element of $\tilde{H}^1(0, \pi)$. Consequently,

$$C_{1,0}(0, \pi) = 1. \quad (4)$$

4 Strict loss at order two

For $u \in \tilde{H}^2(0, \pi)$, the Dirichlet inequality and the mean-zero Wirtinger inequality applied to u' give

$$\|u\|_2 \leq \|u'\|_2 \leq \|u''\|_2. \quad (5)$$

Indeed, u has zero endpoint values and $\int_0^\pi u'(x) dx = u(\pi) - u(0) = 0$. Thus $C_{2,0}(0, \pi) \leq 1$.

We need strictness, because the conjecture concerns the optimal constant. The supremum

$$C_{2,0}(0, \pi) = \sup_{0 \neq u \in \tilde{H}^2(0, \pi)} \frac{\|u\|_2}{\|u''\|_2}$$

is attained. To see this, normalize a maximizing sequence by $\|u_j''\|_2 = 1$. The two inequalities in (5) bound it in H^2 . A weakly convergent subsequence converges strongly in L^2 by Rellich compactness, and the closed trace conditions persist. Weak lower semicontinuity, followed by a rescaling if necessary, yields a maximizer.

Suppose now that $C_{2,0}(0, \pi) = 1$, and let u be a nonzero maximizer. Then equality holds throughout (5). Equality in the first, sharp Dirichlet inequality forces

$$u(x) = A \sin x.$$

But $u \in \tilde{H}^2(0, \pi)$ also has $u'(0) = 0$, so $A = 0$, a contradiction. Hence

$$C_{2,0}(0, \pi) < 1. \quad (6)$$

Theorem 1. *Conjecture 5.8 is false. For $\Omega = (0, \pi)$ and $p = 2$,*

$$C_{2,0}(\Omega) < 1 = C_{1,0}(\Omega)^2,$$

whereas (2) with $(r, z, t, s) = (1, 0, 2, 0)$ predicts equality.

5 Mechanism, verification, and novelty

The conjectured interpolation value remains a valid upper bound; it simply need not be optimal. The obstruction is the changing optimization domain. Raising the zero-extension Sobolev order from one to two imposes the additional normal trace $u' = 0$, excluding the order-one extremizer. After rescaling, the same argument gives a counterexample on every bounded interval.

For orientation only, the order-two extremizer is the first clamped-beam mode: $C_{2,0}(0, \pi) = \pi^2/\beta_1^2 \approx 0.441$, where $\beta_1 \approx 4.730$ is the first positive root of $\cos \beta \cosh \beta = 1$. This numerical identification is not used in the proof; the accompanying script checks it.

The 2023 journal version retains Conjecture 5.8. A bounded search using its exact title and wording, the arXiv identifier, and later papers citing the source found no resolution or matching interval counterexample. Independent mathematical and literature review is nevertheless recommended.

References

- [1] J. Railo and P. Zimmermann, *Fractional Calderón problems and Poincaré inequalities on unbounded domains*, J. Spectr. Theory 13 (2023), 63–131, Conjecture 5.8. <https://doi.org/10.4171/JST/444>